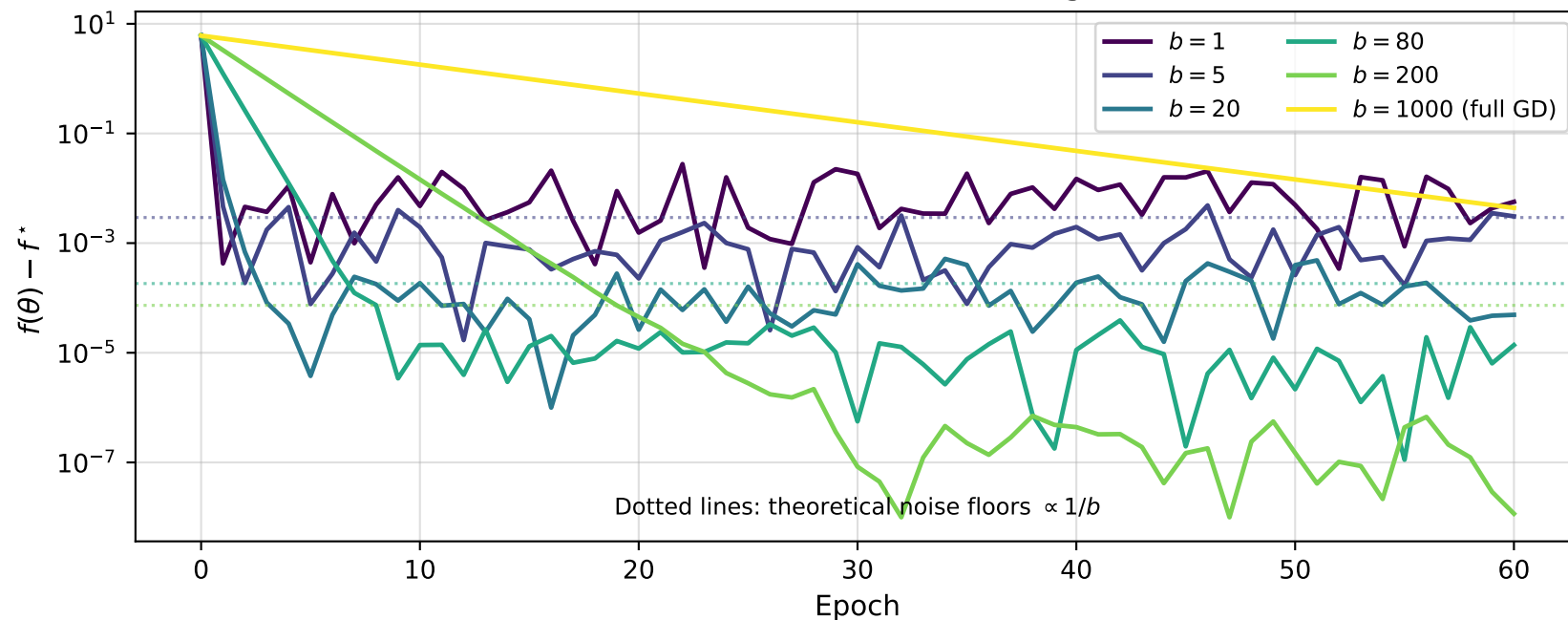
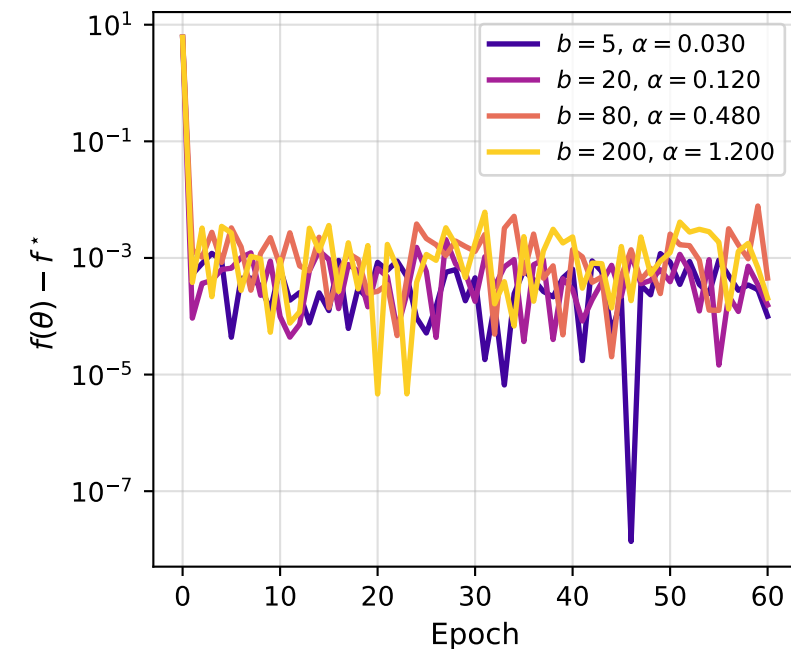


Q2: Mini-Batch SGD --- Batch Size, Noise Floor, and the Linear Scaling Rule

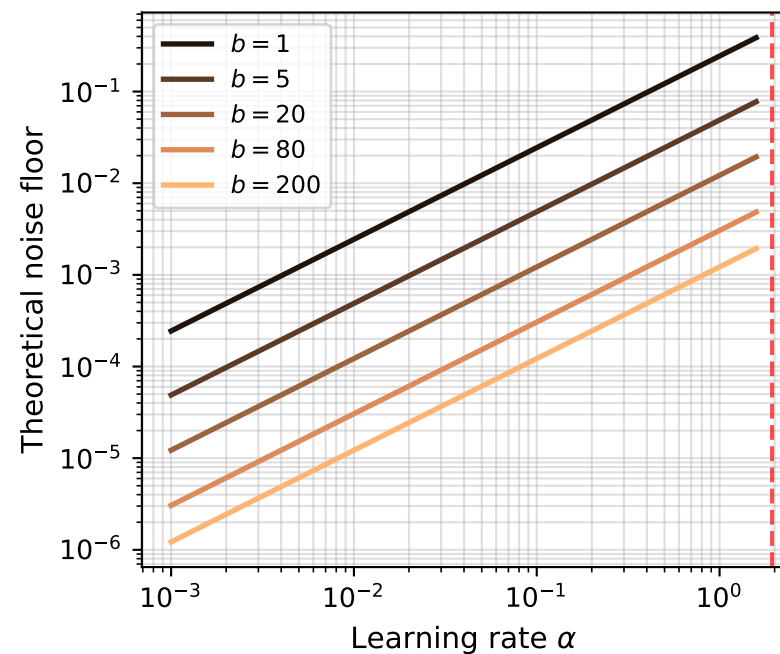
(a) Fixed $\alpha = 0.06$: effect of batch size on convergence and noise floor



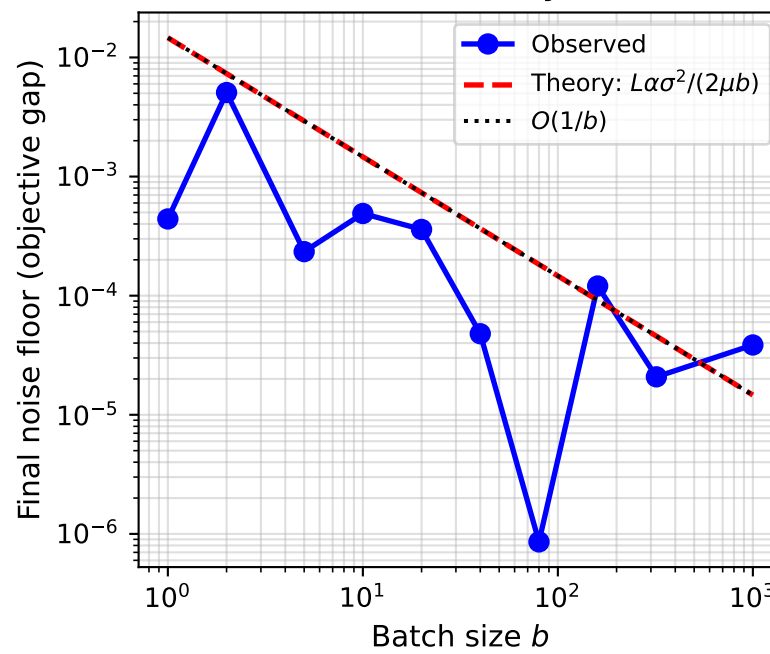
**(b) Linear scaling rule:
 $\alpha/b = 0.0060 = \text{const.}$**



**(c) Noise floor $\propto \alpha/b$
(theoretical, $\kappa = 1.1$)**



(d) Noise floor $\propto 1/b$: theory vs observation



**(e) Convergence speed vs batch size
(fixed $\alpha = 0.06$, large $b \rightarrow$ fewer updates)**

